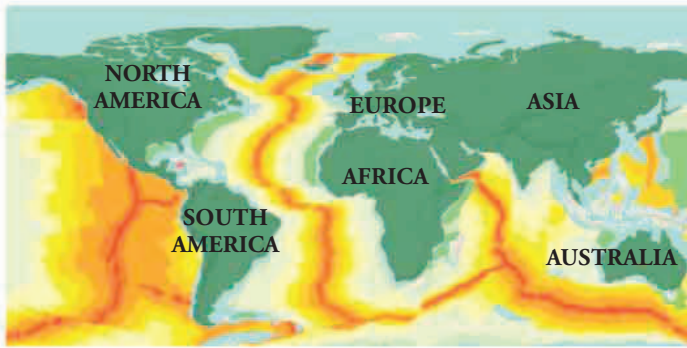


RECYCLED ROCK

Scientists have sampled the rocks of the ocean floors and measured their age. The data shows that the youngest rocks are at the mid-ocean ridges (shaded in red), and that they get older the further they are from the ridge. This proves that the rocks are forming at the ridges and gradually moving away from them. Some of the oldest ocean-floor rocks are being dragged into the subduction zones beneath deep ocean trenches, where they are melted and recycled.

EARTHQUAKE ZONES

The relentless movements of Earth's crust that reshape oceans and move continents also trigger countless earthquakes. Many are felt on land, and sometimes have catastrophic effects. But many more occur beneath the oceans, in the regions where ocean floors are being created at mid-ocean ridges, or destroyed in subduction zones. As a result, the locations of these earthquakes form lines that follow the network of ocean-floor ridges and trenches.



▲ TREMBLING EARTH

The red dots on this map mark the sites of all the earthquakes detected over the last 50 years. The mid-ocean earthquakes form a pattern that matches that of the youngest ocean-floor rocks.



▲ 180 MILLION YEARS AGO

During the Jurassic period, the supercontinent split in two, and the outline of North America began to appear.



▲ 66 MILLION YEARS AGO

By the end of the age of dinosaurs, the Atlantic Ocean had opened up, pushing America away from Europe and Africa.

